



Derived Relational Responding and Intelligence: Assessing the Relationship Between the PEAK-E Pre-assessment and IQ with Individuals with Autism and Related Disabilities

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Abstract

The present study evaluated the relationship between derived relational responding demonstrated by 64 individuals with developmental or intellectual disabilities (90.63% of which had an autism diagnosis) and their corresponding performance on standardized IQ assessments. Participants' relational abilities were assessed using the *Promoting the Emergence of Advanced Knowledge Equivalence Pre-Assessment* (PEAK-E-PA), and IQ was assessed using either the WISC-IV Short Form assessment or the WPPSI-III Short Form assessment. The data indicated a strong, significant correlation between participant scores on the PEAK-E-PA and IQ both in terms of raw IQ ($r = .843, p < .01$), as well as full scale IQ ($r = .804, p < .01$), which is further analyzed in terms of each of the four subtests of the PEAK-E-PA (reflexivity, symmetry, transitivity, and equivalence). The obtained results have implications for a behavior analytic understanding of factors that relate to the development of intelligent behavior in individuals with disabilities.

Keywords Autism · Intelligence · Relational frame theory · Stimulus equivalence

The field of differential psychology has firmly established that people who perform well on some tasks are more likely to perform well on other tasks (see Deary, Penke, & Johnson, 2010), a phenomenon that has spawned over a century of research on intelligence quotient (IQ) testing (Sternberg, Grigorenko, & Bundy, 2001). IQ tests traditionally measure various behaviors hypothesized to indicate “intelligent behavior,” or the ability to think abstractly and adapt to environmental changes (Sternberg & Detterman, 1986; Terman, 1921; Wechsler, 1997). Skills assessed in conventional IQ testing often include general adaptive skills such as problem solving, logical reasoning, vocabulary, and fluency. Although behavioral scientists have philosophically rejected the hypothetical construct of intelligence as a “cause” of intelligent behavior (i.e., behaviors assessed in contemporary intelligence tests) (Schlinger, 1992), attempts have been made to provide a be-

havioral account of the factors that influence intelligent behavior (e.g., Hayes, Barnes-Holmes, & Roche, 2001; Schlinger, 2003), and behavior analytic treatment approaches have demonstrated efficacy in improving participant performance on standardized IQ tests (e.g., Lovaas, 1987; Sallows & Graupner, 2005; Smith, Eikeseth, Klevstrand, & Lovaas, 1997). The behavioral position and related empirical data do not support the premise that intelligence is a trait variable determined by some internal factor that is static over an individual's lifetime; rather there seem to be environmental factors that select intelligent behavior, and these factors can be manipulated to bring intelligent behavior under operant control (Cassidy, Roche, & O'Hora, 2010; Schlinger, 1992). In recent years, there has been an increase in the number of individuals diagnosed with developmental and intellectual disabilities, with an increase of 17.1% for children between the ages of 3 and 17 years from 1997 to 2008, and in which the number of individuals diagnosed with autism has increased by 289.5% (Boyle, Boulet, Schieve, Cohen, Blumberg, Yeargin-Allsopp, & Kogan, 2011). Although an autism diagnosis does not necessarily indicate diminished functioning, Charman, Pickles, Simonoff, Chandler, Loucas, and Baird (2011) found that 55% of children from an autism sample had an IQ of less than

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